

Tutorial-6
ME-101, (Engineering Mechanics, Semester-II)
Date: 04/03/2024 and Time: 7.55 to 8.50 AM

Q1. Predict through calculation whether the homogeneous semi-cylinder and the half-cylindrical shell will remain in the positions shown in Figure 1 or whether they will roll off the lower cylinders (i.e., whether there is a stable equilibrium or an unstable equilibrium for the two conditions).

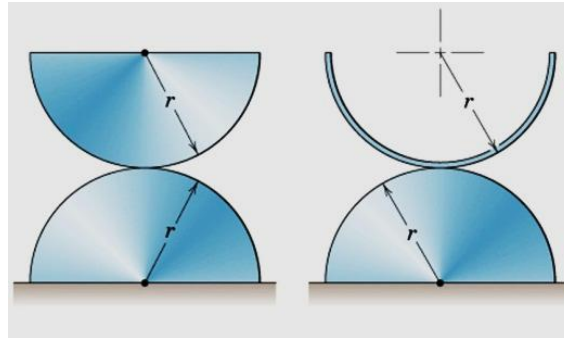


Figure 1

Q2. Determine the second moment of area of the shaded area (refer Figure 2) I_{xy} .

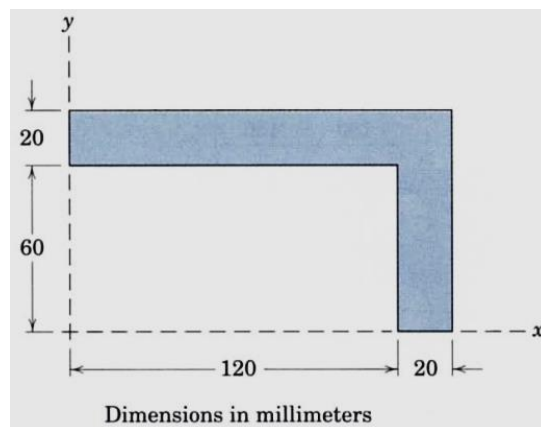


Figure 2

Q3. Determine the angle α which locates the principal axes of inertia through point O for the rectangular area (Figure 3). Construct the Mohr's circle of inertia. Specify the corresponding values of I_{max} and I_{min} .

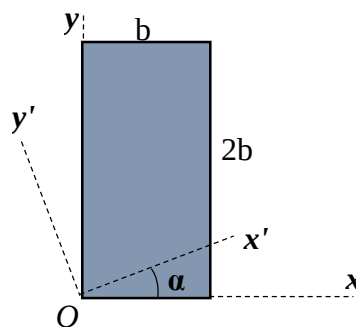


Figure 3

Q4. If equation of parabola is $y = a + bx^2$ (as shown in **Figure 4**) then

- Determine value of a and b [2 marks]
- Determine total shaded area [3 marks]
- Determine total area moment of inertia about y-axis of the shaded area [3 marks]
- Determine the radius of gyration about the y-axis of the shaded area [2 marks]

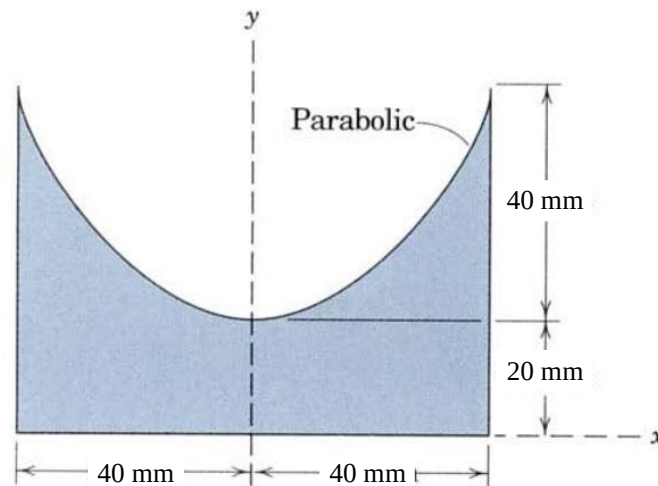


Figure 4

Q5. A homogeneous hemisphere of radius r is placed on an inclined plane as shown **Figure 5**. Assuming that the friction is sufficient to prevent slipping between the hemisphere and the inclined plane, determine

- The largest angle β for which a position of equilibrium exists. [6 Marks]
- The angle θ corresponding to equilibrium, when the angle β is equal to half the value found in part **a**. [4 Marks]

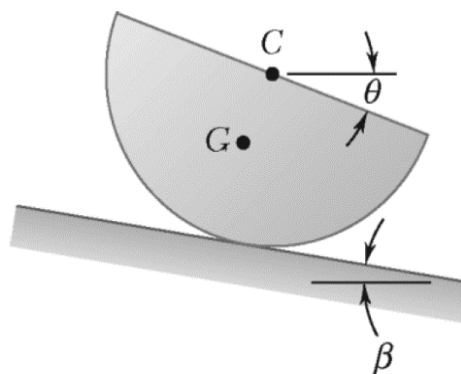


Figure 5